ISR Final Document

IIT BOMBAY INSTITUTE SUSTAINABILITY REPORT 2025
Sustainability Cell

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2. Editorial Note:

We at the Sustainability Cell, IIT Bombay are immensely proud to present you this year's Institute Sustainability Report. The report is the culmination of months of our team's efforts but more-so IIT Bombay's steadfast commitment to sustainability and environmental stewardships. In this edition we have also incorporated IIT Bombay's sustainability policy to increase transparency and consolidate the work being done.

This overview offers a comprehensive look into IIT Bombay's pioneering sustainability journey, including student-led efforts, infrastructural developments, and global recognition for its contributions to the UN Sustainable Development Goals (SDGs).

IIT Bombay stands as a beacon of excellence not just in engineering and technology education, but also in its sustainability ventures. That being said, there is a lot of work yet to be done. I hope that this year's Institute Sustainability Report showcases areas where we as an Institute are doing well, while also reflecting on areas of improvement.

We at the sustainability cell also feel very strongly about creating an eco-friendly, socially just and equitable campus. We hope reading this report brings you along for the ride with us, because the journey is not only ours to traverse.

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3. IIT Bombay at a Glance:

Indian Institute of Technology Bombay (IIT Bombay), located in Powai, Mumbai, was established in 1958 and has since grown into one of India's top ranked engineering institutions. Nestled over 550 acres amidst natural features like lakes, wooded areas, and the adjacent Sanjay Gandhi National Park, the campus environment reflects its ecological ethos.

With 15 departments, numerous interdisciplinary programs, and over 30 research centres, IIT Bombay is not only a hub for cutting-edge academic research but also a model institution in terms of integrating sustainability into its core functions. The campus is home to more than 13,000 students and is recognized globally for its emphasis on education, research, and innovation.

From its early years, IIT Bombay has prioritized sustainability as a central theme in campus planning and institutional policy. It has implemented numerous eco-friendly solutions such as:

- Solar energy systems
- Rainwater harvesting units
- Energy-efficient building designs
- Solid waste management and segregation programs

These initiatives help reduce the institute's carbon footprint while also serving as demonstrative models for other academic and public institutions.

The institute's approach to balancing technological advancement with ecological responsibility reflects a forward-thinking model of development that prioritizes both innovation and the planet.

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4. About the Sustainability Cell and Campus Initiatives:

About the Sustainability Cell

We at the Sustainability Cell at IIT Bombay are a student-led body dedicated to promoting sustainable practices and making the campus a benchmark for environmental responsibility. The Sustainability Cell collaborates with faculty, administration, and students to drive impactful change. It operates through specialized verticals focusing on policy, events, and digital outreach. We also undertake infrastructure improvements, educational events, and data-driven solutions to integrate sustainability across all facets of campus life.

- Leadership and execution by student coordinators and managers, supported by the Hostel Affairs Council.
- Core activity domains: sustainability awareness, waste management, climate action, carbon footprint assessment and reduction, and promoting sustainable resource use.
- *Green Cup* annual competition that encourages hostels to reduce energy and waste through friendly rivalry and behavioral change.
- Collaborations with industry partners, campus bodies, government programs, and NGOs to institutionalize sustainable practices.
- Engagement in sustainability-themed events, competitions, workshops, and community outreach campaigns.
- Focus on embedding sustainability through behavioral change, technological adoption, policy discussions, and transparent communication

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About GESH IITB

The Green Energy and Sustainability Research Hub at IIT Bombay (GESH IITB) is structured as a multidisciplinary, collaborative research and innovation platform that integrates education, research, outreach, and entrepreneurship to address global climate and sustainability challenges. It promotes synergy across various IIT Bombay departments and external stakeholders, including industry, policymakers, and global partners.

- Multidisciplinary Research Hub: GESH IITB focuses on green energy, climate science, clean technology, circular economy, battery and fuel cell technology, biofuels, environmental monitoring, and sustainable manufacturing. It integrates expertise from diverse academic departments like Chemical Engineering, Environmental Science, Civil Engineering, Material Science, Energy Science, and Management.

Core Functional Verticals:

- Collaborative Network and Partnerships: GESH acts as a nexus connecting academia, industry leaders, regulatory bodies, research institutions, startups, and global sustainability platforms. These partnerships enhance resource sharing, enable technology transfer, and promote policy advocacy.
- Infrastructure and Support: The hub operates with dedicated leadership including program managers and consultants, supported by a team of researchers, educators, and administrative staff. It is housed within IIT Bombay's campus facilities, supported by state-of-the-art research infrastructure, labs, and learning environments.
- Focus on Campus as a Living Lab: GESH structurally incorporates IIT Bombay's campus as a testing ground for sustainability innovations, providing hands-on opportunities for students and researchers to develop, pilot, and scale green solutions in real-world contexts

Living Lab Program at IIT Bombay

The Living Lab Program is IIT Bombay's flagship pathway for advancing campus sustainability through real-world experimentation, student-led implementation, and systems-level learning. Led by the IIT Bombay Research Hub for Green Energy and Sustainability (GESH), the program positions the campus itself as a testbed where policies, technologies, and behaviour-change models are piloted before being scaled. It enables the Institute to bridge the gap between research and operations by bringing together faculty, students, campus administration, and external collaborators to co-create practical solutions.

At its core, the Living Lab follows an implementation-first approach. Instead of confining sustainability efforts to awareness campaigns or classroom instruction, the program embeds projects directly within campus systems such as mobility, food services, waste handling, utilities, and student life. This framework allows participants to design interventions that are measurable, replicable, and integrated with real infrastructure. Through this model, the Living Lab fosters institutional learning while nurturing a culture of responsible campus stewardship.

The design of the program encourages broad participation across the IIT Bombay community. Fellows are selected to lead projects and work alongside hostel councils, mess contractors, facility managers, institute bodies, and relevant technical units. The aim is to develop working models that are scalable to other institutions, rather than standalone pilot outcomes. The first cohort focuses on nine projects across four key themes: behavioural change, waste management, water systems, and energy and climate. Together, they establish a foundation for a multi-year platform that aligns student innovation with campus-level sustainability planning.

The Nine Living Lab Projects

Green Cup: Inter-hostel Sustainability Competition

The Green Cup project strengthens campus culture by combining student enthusiasm for inter-hostel competitions with sustainability performance. The initiative creates a structured scoring framework through which hostels are evaluated on sustainable practices such as resource efficiency and waste reduction. Pilot activities such as occupancy sensors and zero-waste days help identify feasible interventions within hostel infrastructure. By using data-backed monitoring and incentives, the project encourages student ownership of environmental responsibility while integrating sustainability into everyday campus life.

General Sustainability Course for First-year Students

This project introduces a campus-specific sustainability module for incoming students to help align individual behaviour with institute goals. Through interactive content, real campus data, and applied examples, the module familiarises first-years with energy use, water conservation, waste handling, and mobility choices. Anchoring this training within student orientation ensures early engagement and shared understanding of sustainable campus norms. The long-term aim is to improve baseline literacy across batches, making campus-wide adoption of sustainable practices easier to sustain.

Sustainable Event Planning Guidelines

With a large number of events held on campus each year, this project addresses the environmental footprint of event-related consumption and waste. The team has developed guidelines that simplify the process of planning resource-efficient events and includes provisions such as a carbon footprint estimation tool and improved procurement choices. It also proposes a no-banner policy based on an analysis of life-cycle impact and operational feasibility. The guidelines serve as a reference for event organisers and create consistent expectations for sustainable campus convening.

Hostel Food Waste Management

This initiative focuses on reducing plate and counter waste generated in hostel messes through targeted behavioural and operational interventions. The project identifies causes of excess food waste and works with stakeholders to design a mess-level management framework. Awareness, monitoring, and feedback loops are used to shift consumption patterns toward more responsible use. The project also lays the foundation for future models that integrate infrastructure, planning, and improved accountability mechanisms.

ReCycle Café: Circular Mobility

This project advances sustainable mobility by refurbishing abandoned or unused bicycles and putting them back into circulation through rentals, low-cost sales, and donations. By promoting reuse rather than new purchases, the initiative supports circular economy principles while improving last-mile connectivity for students. It also enhances accessibility for economically disadvantaged users. The longer-term vision is to create a structured reuse chain that keeps bicycles in active service for as long as possible.

Wet Waste Management: Biogas vs Composting Feasibility Study
This project evaluates IIT Bombay's existing waste treatment infrastructure and compares composting and biogas systems on techno-economic and environmental grounds. The study supports institutional planning by identifying current capacities, projected future demand, and opportunities for system-level optimisation. It also provides evidence for decision-making related to expansion of onsite treatment. The findings help the campus plan more efficiently for long-term waste-to-resource systems.

Water Demand Mapping and Audit
This initiative establishes a baseline understanding of campus water supply, distribution, and wastewater flows. Through GIS-based mapping and auditing of high-consumption zones, the project identifies opportunities for improved conservation and reuse. It also proposes technical inputs such as smart metering and recommendations on potential STP locations. The data collected under this project informs infrastructure planning and supports more efficient water governance.

Carbon Footprint Assessment
This project builds a structured framework for calculating baseline greenhouse gas emissions arising from campus operations. The assessment covers Scope 1 and Scope 2 emissions and also incorporates estimates of the carbon sequestration potential of campus vegetation. The resulting framework enables future carbon reduction planning and provides a template for carbon accounting that can be replicated by other institutions. It creates the foundation for future decarbonisation pathways at IIT Bombay.

Chiller Optimisation and Campus Energy Efficiency
This project targets energy-intensive cooling systems and explores optimisation opportunities through data-driven analysis. It examines performance trends and identifies steps that can improve energy efficiency without compromising comfort. The study also includes complementary assessments of other high-use electrical systems such as lighting and distributed solar generation. The work supports long-term campus energy planning and contributes to operational efficiency.

Student-led and research-driven Sustainability

Some other prominent student-led bodies include:

Team Shunya - Team SHUNYA (Sustainable Housing for an Urbanizing Nation by Young Aspirants) is IIT Bombay's interdisciplinary student team dedicated to designing affordable, net-zero, solar-powered homes tailored for India's urban population. Founded in 2012, the team has earned global recognition through competitions like the Solar Decathlon, recently winning 1st place in the 2024 U.S. Build Challenge for their retrofit housing model “Project Samsara”. Comprising students from across various departments and guided by faculty, SHUNYA blends engineering, architecture, and sustainability to promote green building and climate-resilient housing.

Energy Club - Energy Club at IIT Bombay, is a student-driven interdisciplinary body founded in 2011 to promote energy conservation and sustainable development. Its mission is to transform the campus into a model of “green energy efficiency” while driving innovation in technology, policy, and awareness. The club conducts technical workshops, policy discussions, and awareness campaigns, fostering holistic engagement with energy transition challenges. Key initiatives include “Renewathon”, a national-level challenge on real-world energy problems; “Energize”, an energy-themed campus scavenger hunt; and field visits to renewable energy sites. By combining deep technical insight with community action, the club builds a culture of environmental responsibility and forward-thinking energy solutions.

NSS- NSS IIT Bombay is the institute’s official chapter of the National Service Scheme, run under the Ministry of Youth Affairs & Sports anchored in the values of “Think We, Not Me”. With over 350 volunteers and a 40-member core team, it drives impactful community work through three key verticals: Educational Outreach, Sustainable Social Development, and Green Campus. From teaching underprivileged children and conducting digital literacy drives to leading waste management and environmental clean-up campaigns, NSS blends grassroots action with student leadership. Actively collaborating with NGOs and civic bodies, it offers IITB students hands-on opportunities to create real, sustainable change.

Abhuday - Abhyuday is IIT Bombay’s student-run social body, active from 2013-14, it focuses on civic responsibility and youth-led impact. It organizes awareness drives, cleanups, and its flagship Social Fest, while also running national programs like the CR Program and Social Fellowships. Fully student led, it partners with NGOs and corporates to drive large-scale change.

Enactus - Enactus IIT Bombay is a student-led social entrepreneurship team that develops sustainable, market-ready solutions to real-world challenges. Their key projects include “SusPak”, which creates compostable packaging from agricultural waste to replace single-use plastics; “Project Ragini”, empowering rural women through ragi-based micro-enterprises; and “Project Atithi”, promoting eco-tourism in villages. By combining innovation with social impact, Enactus IITB fosters scalable change through community-driven, entrepreneurial action.

Zero Waste - Team Zero-Waste is a student-driven initiative launched in 2024 under the Tata Centre for Technology and Design, aiming to make the campus truly zero-waste-to-landfill. The team works across three fronts - technology, policy, and awareness, to design smart waste solutions, reduce single-use materials, and promote sustainable behavior. From deploying automated segregation systems to running plastic-free campaigns and circular event plans, they blend innovation with action. Notably, their intervention at the IITB Half Marathon replaced 13,000 plastic bottles with clay kulhads, cutting emissions and water waste while supporting local artisans.

These platforms cultivate a culture of hands-on learning and real-world problem-solving in sustainability.

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5. Executive Summary

IIT Bombay has advanced multiple transformative initiatives that demonstrate its leadership in environmental stewardship and innovation.The campus has strengthened its green infrastructure through solar power plants, biogas facilities processing residential food waste, rainwater harvesting and greywater recycling systems, and GRIHA-rated energy-efficient buildings. Sustainable mobility has been enhanced with electric shuttles, e-rickshaws, cycling lanes, and expanded pedestrian pathways. Waste management has become a benchmark practice, with campus-wide segregation, composting, biogas generation, and recycling systems for plastics, e-waste, and biomedical materials.

On the academic front, IIT Bombay is planning to introduce a **Minor in Sustainability**, offering interdisciplinary courses that bridge policy, economics, and technology to prepare students for the growing green economy. In addition to this initiatives such as the **Living Lab Fellowship**, **Greenovate Challenge**, and **Green Cup 2.0** have fostered experiential learning and student-led innovation in areas of energy, waste, water, and climate action.

Research excellence continues through the **Green Energy and Sustainability Hub (GESH)**, which drives innovation in renewable energy, sustainable manufacturing, circular economy, and climate science. Faculty-led projects have yielded pioneering solutions in water treatment, wetland-based sewage systems, and carbon footprint monitoring, while industry collaborations and global partnerships amplify impact and knowledge exchange.

Student-led bodies, **Team Shunya, NSS, Abhyuday, Enactus, Zero Waste, and the Energy Club**, have expanded IIT Bombay's sustainability footprint through award-winning housing prototypes, large-scale plantation drives, zero-waste initiatives, and nationwide outreach programs.

These efforts have translated into international recognition, with IIT Bombay making a significant leap in the **QS Sustainability Rankings**, reinforcing its position as a leading model of higher education sustainability in India. The report reflects a campus-wide culture of innovation, responsibility, and collaboration, positioning IIT Bombay at the forefront of advancing the UN Sustainable Development Goals while shaping a resilient, net-zero future.

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6. The Sustainability Policy

In 2025, IIT Bombay released its *Sustainability Policy*, setting out the institute's goals and commitments toward a more sustainable future. In this year's Sustainability Report, we aim to track and present the progress made on each of the activities outlined in the policy, ensuring coherence, transparency, and accountability throughout.

Each goal from the policy is categorized as *Completed*, *In Progress*, or *Not Completed*, along with detailed notes on the corresponding initiatives. This approach not only highlights the institute's achievements but also provides a clear picture of IIT Bombay's ongoing journey toward building a resilient, net-zero, and sustainably governed campus.

7. Environmental Awareness, Education and Research

Goals	**Progress Icon**	**Details**
Some of the awareness activities include collection of plastics and glass from various locations in the campus, zero waste days in messes, display of posters, plays/ skits on the roads or hostels.	Completed	NSS student volunteers conduct drives such as cleanliness campaigns, plastic collection, poster displays, and zero waste days. Zero waste days encourage residents to avoid disposables and use only reusable alternatives.
IIT Bombay is determined to promote environmental education among kids via competitions, cleaning drives and plantation activities.	Completed	Student bodies organize various competitions, cleaning drives, and plantation activities throughout the year to promote active participation in sustainability.
The professionals working in diverse fields or similar disciplines to improve awareness at the National level are also trained through short courses and hands-on training programmes.	in progress	Departments and centers are developing sustainability-oriented short-term courses, such as EV charging training for professionals, to build technical knowledge.
Competitions among hostels will be planned particularly for food waste minimization.	Completed	The Sustainability Cell organizes Green Cup, an institute-wide competition where hostels earn points for sustainable practices, including minimizing food waste.
Environmental awareness programmes will be organized for the campus residents to adopt sustainable consumption practices	Completed	NSS and the Public Relations Office run awareness campaigns and workshops to encourage residents to adopt sustainable lifestyles.
Train professionals in sustainability & circular economy.	Completed	GESH IITB conducts a 3-day program on ESG, Circular Economy strategies, and life-cycle thinking to train professionals in sustainability leadership.

<u>Proposed Minor in Sustainability</u>

Why - India is set to create 35 million green jobs by 2047, with the green economy growing to $1 trillion by 2030, driven by renewables, electric vehicles, green construction, and waste management. Sustainability skills are becoming essential, with hiring demand in green sectors rising 15–20% annually. India's renewable energy capacity has tripled in the last decade to over 230 GW, boosting demand for skilled professionals. Leading global university rankings now include sustainability education, impacting reputation and student choices. For IIT Bombay, a Sustainability Minor addresses real job market trends, enhances institutional standing, and meets evolving student and employer priorities.

Who - Considering sustainability is a diverse and interdisciplinary field, multiple courses will be offered from various departments.

a. These mainly include Climate Studies, Environmental Engg, Energy Engg, Chemical Engg, Civil Engg, and Policy studies.
b. These courses span from holistic - systems level approach (Like Introduction to Urban Design, Energy Policy and Planning) to specific - technical aspects of sustainability (Waste to Energy, Materials for Sustainable Development)

What - The Sustainability Minor integrates two complementary tracks:

1. Policy, Economics & Management:
    a. Covers environmental governance (ESG, water and energy policy, urban design, sustainable finance) through management, policy, entrepreneurship, and humanities.
    b. Builds foundational knowledge of regulatory frameworks, market incentives, and strategic planning essential for sustainable solutions.
2. Engineering & Technology:
    a. Includes climate science, renewable energy (solar, hydrogen, batteries), sustainable construction, water and waste management, and advanced materials, leveraging IITB's engineering expertise.
    b. Emphasizes technical innovation and problem-solving for infrastructure and clean technologies.

By combining these perspectives, students learn to develop technically sound, economically feasible, and policy-compatible solutions. With global commitments to halve carbon emissions by 2030, this collaboration prepares them for the rising green jobs requiring multidisciplinary expertise and equips graduates to effectively translate sustainability ideas into scalable, impactful actions.

Final Benefits -

1. Develop systems thinking and sustainability leadership.
2. Equip students with tools to analyze and solve environmental, economic, and social challenges.
3. Prepare graduates for green jobs, innovative entrepreneurship, and effective policy or engineering roles in the rapidly growing sustainability sector
4. Make this into an Infographic - Design Team

Research Labs in Spotlight

Research Title	Professor & Team	Domain	Essence of Research
Zonal storage water supply system with hydraulic isolation structure (ZS-HIS)	Prof. Pradip Kalbar, Centre for Urban Science & Engineering	Urban Water Management	Developed a system to manage zonal water supply, minimize wastage, ensure equal distribution, and improve the efficiency and reliability of urban water supply.
Integrated Wetland Technology	Prof. Anil Kumar Dikshit, Environmental Department	Wastewater Treatment, Low-Carbon Tech	Uses wetlands for affordable, low-carbon, energy-efficient wastewater treatment, treating organic and inorganic pollutants, with applications in urban and peri-urban sanitation.
SafeSan: Safe sanitation system	Prof. Anil Kumar Dikshit	Sanitation, Wastewater	Developed a cost-effective, efficient, and safe on-site sanitation and treatment system that removes organics and pathogens, increasing access to sanitation and reducing environmental pollution.
Photocatalytic Membrane Reactor for Water Treatment	Prof. Jayesh Bellare	Water Treatment	Uses photocatalytic membranes for removing dyes and pharmaceuticals from water, with a low-cost, high-efficiency design for tackling complex pollutants.
High Flux Anti-fouling HFM for Water Treatment Application	Prof. Jayesh Bellare	Membrane Tech, Water Treatment	Developed advanced, anti-fouling, high-flux membranes to address fouling in water filtration, promising higher efficiency and cost reductions.
Mobile Water Treatment Plant (mWTP) for Emergency Situations	Prof. Anil Kumar Dikshit & Team	Emergency Water Supply	Portable plant to provide safe drinking water in emergencies or remote areas, supporting disaster relief and resilience.
Green Sewage Treatment Plant Based on Integrated Wetland Tech	Prof. Anil Kumar Dikshit	Sewage Treatment, Wetland Systems	Sustainable STP using wetland-based treatment, reducing costs and enabling eco-friendly wastewater recycling for urban and rural communities.
U-bent optical fiber sensors	Prof. Soumyo Mukherji	Sensors, Environmental Monitoring	Developed sensitive, affordable sensors to detect pollutants and pathogens in water, supporting real-time water quality monitoring and health.
Smart ultrasonic water meter with AMI	Prof. P. S. V. Nataraj	Water Metering, IoT	Created a smart, low-cost, ultrasonic water meter to monitor and analyze

			water usage for improving management and conservation.

Faculty Research in Sustainability Domains

Faculty across IIT Bombay are actively contributing to sustainability through diverse areas of research. Their work spans renewable energy, water management, air quality, climate change, and sustainability policy, shaping both technology and governance solutions for a more sustainable future.

Renewables & Energy Systems

Faculty are pioneering research in renewable technologies and sustainable energy systems. Work in this domain includes solar energy innovations, carbon capture, clean energy pathways, and energy transitions.

Some of the faculty contributing include: Prof. Munish Chandel, Prof. V.S. Vamsi Botlaguduru, Prof. Vishal Dixit, Prof. Anjali Sharma, Prof. Shireesh B. Kedare, Prof. Santanu Bandyopadhyay, Prof. Rangan Banerjee, Prof. Chetan Singh Solanki, Prof. Sandeep Kumar, Prof. Shaibal K. Sarkar, Prof. Srinivas Seethamraju, Prof. Suneet Singh, Prof. Suryanarayana Doolla, Prof. Venkatasailanathan Ramadesigan, Prof. Anish Modi, Prof. D. Venkatramanan, Prof. Gurubalan Annadurai, Prof. Karthik Sasihithlu, Prof. Prakash Chandra Ghosh, Prof. Rajesh Gupta, Prof. Sankara Sarma V. Tatiparti, Prof. Satish B. Agnihotri, Prof. A.W. Date, Prof. Uday N. Gaitonde, Prof. Yogendra Shastri, Prof. Sanjay Mahajani, Prof. B.A. Vijayasree, Prof. Zakir Hussain Rather, Prof. Manoj Neergat, Prof. Pratibha Sharma, Prof. Sagar Mitra, Prof. Siddavatam Ravi Prakash Reddy, Prof. Balasubramaniam Kavaipatti, Prof. Dayadeep Monder.

Water & Wastewater

Research in this area addresses sustainable water management, treatment technologies, and hydrological systems for safe water access and pollution control.

Some of the faculty contributing include: Prof. Sanjeev Chaudhari, Prof. Anil Kumar Dikshit, Prof. Anurag Garg, Prof. Subhankar Karmakar, Prof. Suparna Mukherji, Prof. Swatantra Pratap Singh, Prof. Indrajit Chakraborty, Prof. Tabish Nawaz, Prof. Amritanshu Shriwastav, Prof. Renuka Verma, Prof. Lalit Kumar, Prof. N.C. Narayanan, Prof. Bakul Rao, Prof. Pennan Chinnasamy, Prof. Himanshu Kulkarni.

Air Pollution & Environment

Faculty are advancing knowledge on sources and impacts of air pollution, while developing mitigation strategies to safeguard public health and ecosystems.

Some of the faculty contributing include: Prof. Manoranjan Sahu, Prof. Harish Phuleria, Prof. Abhishek Chakraborty, Prof. Virendra Sethi, Prof. Asish Kumar Sarangi.

Climate & Carbon Management

This cluster focuses on climate science, carbon reduction pathways, and adaptation strategies. Research includes climate modelling, socio-economic impacts of climate change, and carbon management.

Some of the faculty contributing include: Prof. Srinidhi Balasubramanian, Prof. S. Ravichandran, Prof. Vikram Vishal, Prof. Angshuman Modak, Prof. Akshaya Nikumbh, Prof. Suyash Bire, Prof. Jaideep Joshi, Prof. D. Parthasarathy, Prof. Raghu Murtugudde.

Sustainability & Policy

Faculty are engaged in the social, economic, and policy dimensions of sustainability. Their work explores governance, circular economies, sustainable development frameworks, and integration of science with public policy.

Some of the faculty contributing include: Prof. Radha Kale, Prof. Om P. Damani, Prof. Pankaj Sekhsaria, Prof. Pradip Kalbar.

These lists are illustrative and highlight selected faculty members working in each domain. They are not intended to be exhaustive.

Event:
Energize
Body:
Energy Club
Details:
The Energy Club organized Energize on 3rd October 2024 , bringing together students for an evening of engaging activities focused on sustainability and energy awareness. Through cryptic hunts, crosswords, and interactive games, participants explored themes like energy transition, climate action, and the SDGs in a fun, team-based format. The event aimed to promote environmental consciousness while offering a much-needed break post-midsems, concluding with a surprise segment that reinforced the club’s commitment to combining learning with meaningful impact.

Event:
Sustainability Summer School
Body:
Career Cell & Other Bodies
Details:
Career Cell in association with various student bodies launched seven summer courses under the Learners Space Program. These included 'Sustainable Policy 101' by Sustainability cell, 'Sustainability Research & Policy'; 'Zero Waste Vision' by Team Zero Waste, 'Sustainable Economics: Markets, Policies, and the Planet' by the Economics Association, The 'Carbon Code: Trading, Tracking & Transforming Emissions'; 'Solar Energy & The Global Trend' by the Energy Club, Sustainable Buildings by Team Shunya. These courses covered various aspects of sustainability, offering real world insights into challenges and solutions.

Event:
TS Trainee Program
Body:
Team Shunya
Details:
Team Shunya launched the TS Trainee Program, a hands-on training initiative for students interested in sustainability and green technologies. It covered core concepts like sustainable design, energy efficiency, and environmental stewardship. Participants engaged in group discussions, energy audits, life cycle assessments, and low-carbon design exercises. With a focus on mentorship and real-world problem-solving, the program built key skills for campus sustainability projects and fostered a community of environmentally conscious student leaders.

Event:
Expert Talk: The Complexity of Product Sustainability
Body:
GESH
Details:
The expert talk on "The Complexity of Product Sustainability" explored what it really takes to make products more environmentally responsible. Rather than offering simple solutions, the speaker emphasized a life-cycle approach considering every stage from raw materials to disposal. They highlighted the trade-offs involved in sustainable design, like balancing cost with carbon impact or choosing between recycled and durable materials.

Event:
Advitiya - Circular Economy Speaker Session, CSR Conclave
Body:
Abhyuday
Details:

The session also touched on concepts like circular design and creating products people want to keep for longer. Overall, it was a practical and honest look at how sustainability in product design is rarely black and white, and why thoughtful, context-driven choices matter most.

Event:
Formal Course: Sustainability – Theory to Action
Body:
GESH
Details:
The GESH–SJMSOM course “Sustainability – Theory to Action” was launched to bridge the gap between academic knowledge and practical implementation of sustainability. Its curriculum covered green energy systems, circular economy models, climate policy, and stakeholder engagement, complemented with case studies, group projects, and experiential exercises to foster critical thinking and innovative problem-solving. The course attracted a diverse cohort of students and professionals, creating space for interdisciplinary dialogue and collaboration. A key highlight was its integration with the Living Lab at IIT Bombay, where it was made a compulsory component for all fellows. Through project-based work, fellows were able to scale their Living Lab initiatives, applying classroom learning directly to campus-based experiments in energy, water, waste, and climate action. The course featured guest lectures by leading faculty including Prof. Yogendra Shastri and Prof. Pankaj Jhunja, while Prof. Trupti Mishra played a dual role: teaching modules on climate economics and policy, and mentoring fellows to align their sustainability pilots with broader research and governance frameworks.

Event:
Seminar on Sustainable Chemical Manufacturing
Body:
GESH
Details:
The Sustainable Chemical Manufacturing Seminar, hosted by IIT Bombay’s GESH and IIChE (Indian Institute of Chemical Engineers) Mumbai Regional Centre, convened academic, industry, and policy experts to explore innovations in green chemistry and low-impact production. With keynote speaker Prof. A. B. Pandit and panels featuring leaders from Reliance, UPL, GRP, ICT, and more, the seminar addressed recycling, biomass utilization, CO2 conversion, clean hydrogen, and sustainable fertilizer production. By showcasing practical solutions and forward-looking strategies, the event fostered cross-sector collaboration toward a more sustainable chemical industry.

Event:
Sustainium Case Competition
Body:
Sustainability Cell
Details:
The Sustainability Cell collaborated with Tarutium Global Consulting to organize a worldwide case competition focusing on innovative waste management solutions. Over 500 participants

from across 5+ nations and 100+ universities competed for cash prizes and pre-placement interviews (PPIs).

Event:

Awareness Programs: No Planet B Podcast series

Body:

Energy Club

Details:

A podcast series released on the Spotify account of the energy club. This consisted of a total of 3 podcasts with some eminent speakers from around the world whose work is inlined with the study of climate change. This was planned to focus on the issue of climate change and how it can be controlled by shifting to green energy. Speakers - Abhimanyu Dhariwal (ElectrifyIt CEO, IITB'15), Devansh Timbadia (PM, Oorja), Dr Ajay Mathus (DIrector General, ISA)

Event:

Sustain 4.0 and Sustain 5.0

Body:

Team Shunya

Details:

SUSTAIN 4.0 and 5.0, hosted by Team SHUNYA at IIT Bombay, were immersive sustainability events combining expert talks, panel discussions, and hands-on workshops like EcoCraft and computational design. A key feature was the Startup Summit, which highlighted early-stage green startups and encouraged discussions around climate entrepreneurship. The events showcased innovations like solar thermal systems, eco-packaging, and green financing tools and a standout feature was the SHUNYA House Tour, presenting Project Vivaan; a solar-powered, net-positive home with passive design and rainwater harvesting. Together, these events advanced campus sustainability culture and climate innovation.

Event:

3-Day Executive Programme on Sustainability & Circular Economy

Body:

GESH

Details:

The GESH IITB initiative introduced the 3-Day Executive Programme on Sustainability & Circular Economy, designed to empower leaders and changemakers with actionable skills. The program covered ESG, life-cycle thinking, circular economy strategies, and impact assessment methods, aimed at future-proofing businesses. Participants learned from IIT Bombay faculty through real-world case studies and hands-on sessions, focusing on building strategies to drive sustainability and innovation. The program also facilitated networking with industry pioneers, policymakers, leaders, and academics. Certified learning commences on 18th September.

Event:

Greenovate 2025 Challenge

Body:
GESH
Details:
The Greenovate 2025 Challenge, hosted by GESH IITB on World Environment Day, 5 June 2025 at VMCC, IIT Bombay, was an idea pitch competition showcasing student-led sustainability innovations. Presented in collaboration with the Centre for Climate Studies, the Desai Sethi School of Entrepreneurship, and the IIC, the event featured over twenty breakthrough concepts across clean tech, waste management, water treatment, and carbon capture. Among the winning teams were SusPak with compostable trays and GreyCycle with greywater recycling, followed by runners up Ecoard Board with water hyacinth packaging and ScrapXchange with peer to peer scrap trading.

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8. Sustainable transport

Goals	Progress Icon	Details
The increasing adoption of sustainable transport (such as walking, bicycles etc.) within the campus will contribute towards achieving zero carbon emissions.	In Progress	EV buggies and e-rickshaws are introduced on campus, alongside initiatives like cycle rentals and a Recycle Café, to encourage low-carbon mobility.
The institute will aim to eliminate gasoline and diesel operated vehicles in a phase-wise manner.	In Progress	The institute is restricting rickshaws and other motorized vehicles while installing EV chargers across locations to promote the shift to electric mobility.
Public transport at the campus will be strengthened by taking suitable measures.	In Progress	Shuttle services using EV buggies and e-rickshaws are being expanded to improve connectivity within the campus
Apart from this, bicycle lanes will be constructed to allow safe travel and encourage use of unpowered mode of transport.	In Progress	Dedicated bicycle lanes and road extensions (e.g., near H14 to H6) are being developed to ensure safe cycling.
Additional footpaths are being constructed for pedestrians for easy access to the academic area from hostels.	In Progress	Additional footpaths, such as the Infinity corridor, are being built to improve walkability and safe access between hostels and academic areas.

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9. Water conservation and wastewater recycling

Goals	Progress Icon	Details
Rainwater harvesting programmes will be executed. Some of the examples may include construction of check dams (at feasible locations) at storm water drains, groundwater recharging etc.	In progress	Rainwater Harvesting system is installed at Rahul Bajaj building.
Grey water from residential buildings will be treated and recycled as much as possible.	In Progress	Systems at hostels like H10 and Ananta Building treat wastewater for reuse, though some installations currently need operational improvement.
Provisions will be made to reduce water consumption. Some examples may include installation of low water flush tanks and water less urinals in rest rooms, regular monitoring of water consumption	In Progress	Tap aerators and waterless urinals are installed across restrooms to reduce water usage significantly.
The supply pipes will be maintained and repaired to eliminate loss of water due to leakages.	In Progress	A proposal to install water meters across campus has been passed to enable better monitoring.
Water audits will be performed once in a year to identify the limitations and overcome the issues.	In Progress	Annual audits will be conducted once meters are operational to track usage, identify leakages, and recommend corrective actions.

Event:
Liquid Waste Segregation Bins
Body:
Sustainability Cell
Details:
IIT Bombay's liquid segregation bins initiative by the Sustainability Cell, mandates campus wide at-source segregation of waste into dry (recyclables) and wet (liquid/biodegradable) fractions in all hostels and residences. Specially designed bins capture liquid kitchen waste separately from solid wet waste: kitchen staff first pour liquids such as leftover curries or gravies into the designated liquid compartment, while food scraps go into the solid section. Liquids are routed directly to appropriate composting or biogas plants, reducing excess moisture in the solid waste

stream. This segregation dramatically improves the efficiency of composting and biogas production by ensuring optimal moisture content, enhancing gas yields, and reducing operational issues like leachate. Pilots in Hostel 3 showed measurable improvement in biogas plant performance and reduced contamination in recyclables. Ongoing feedback from students and staff led to ergonomic design tweaks and simplified usage, paving the way for smooth campus-wide adoption and setting a benchmark for institutional waste management.

Event:
Green Plumbing Project
Body:
Team Shunya
Details:
This project incorporated water-saving fixtures, greywater recycling (collecting and treating wastewater from sources like showers, sinks, and washing machines to reuse it for non-potable purposes like toilet flushing and irrigation), rainwater harvesting, and solar-powered hot water into sustainable housing prototypes. The initiative achieved up to 82% water savings compared to conventional houses. By promoting circular economy practices and utilizing eco-friendly materials, the project showcased scalable and affordable plumbing solutions, inspiring sustainable urban housing approaches and raising awareness about water efficiency and environmental responsibility in India.
Moreover, tap faucet aerators were proposed to be installed in hostel washrooms. Faucet aerators work by mixing air into the water stream as it flows from the faucet, creating a softer, more even flow while reducing water usage and splashing. This can be a promising solution to reduce water consumption in hostels, without compromising hygine quality.

Event:
AC Optimization in Lecture Hall Complex
Body:
Sustainability Cell
Details:
To reduce AC-related energy losses in IIT Bombay's Lecture Halls, a pilot replaced heavy, inefficient doors with lightweight marine plywood doors equipped with automatic springs and fire-safety windows. The upgrade helped retain cooled air and reduce electricity use. Based on its success, the plan is to expand this across all LHCs, pairing it with spring installations or full door replacements as needed. Post-upgrade, student and faculty feedback will guide chilled water supply adjustments to align AC settings with comfort levels, ensuring both efficiency and comfort.

Event:
Liquid Waste Segregation Bins
Body:
Sustainability Cell
Details:

IIT Bombay's liquid segregation bins initiative by the Sustainability Cell, mandates campus wide at-source segregation of waste into dry (recyclables) and wet (liquid/biodegradable) fractions in all hostels and residences. Specially designed bins capture liquid kitchen waste separately from solid wet waste: kitchen staff first pour liquids such as leftover curries or gravies into the designated liquid compartment, while food scraps go into the solid section. Liquids are routed directly to appropriate composting or biogas plants, reducing excess moisture in the solid waste stream. This segregation dramatically improves the efficiency of composting and biogas production by ensuring optimal moisture content, enhancing gas yields, and reducing operational issues like leachate. Pilots in Hostel 3 showed measurable improvement in biogas plant performance and reduced contamination in recyclables. Ongoing feedback from students and staff led to ergonomic design tweaks and simplified usage, paving the way for smooth campus-wide adoption and setting a benchmark for institutional waste management.

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10. Solid waste treatment and recycling

Goals	Progress Icon	Details
At source, waste segregation into dry and wet waste fractions is mandatory.	Completed	Dustbins across campus are converted into a dry/wet waste system, making segregation at source mandatory.
All the wet biodegradable waste produced from houses and hostels is processed using appropriate biological processes	Completed	Hostel and residential food waste is processed through a biogas plant (capacity up to 2 tonnes/day) and compost pits, with plans to expand further
The hostels' messes have to reduce food wastage by taking suitable measures.	Completed	Messes run campaigns such as Swachhata Pakhwada, posters, and zero waste days to raise awareness and reduce wastage.
The not-readily biodegradable woody material may be processed to form pellets within the campus if it is found feasible.	Completed	A pelletizing machine processes woody waste into pellets, which are gasified to produce steam used in staff canteens.
The recyclable dry household waste will be recycled either in-house or through an authorised recycler.	Implemented	Dry household recyclables such as glass, paper, and plastics are collected and sent for recycling through authorized vendors.
Sanitary waste will be collected separately and disposed off through an	Implemented	Sanitary waste is collected separately and disposed of safely through an authorized biomedical waste collection agency as per

authorized biomedical waste collection agency.		rules.
Single use plastics is banned in the campus. The buyers are encouraged to carry recyclable bags.	In Progress	Single-use plastics are banned on campus, and buyers are encouraged to avoid them, with awareness campaigns reinforcing this.
Waste plastic will be collected and recycled in the proper manner.	In Progress	Plastic waste is collected in proper bins, and the Sustainability Cell is finalizing reverse vending machines (RVMs) for better recycling.
Biomedical waste from hospitals is being handled and stored as per Biomedical Waste Management Rules (2016) before disposal through an official agency.	Completed	Implemented
E-waste and discarded scrap material (such as old equipment, and their parts, unusable furniture etc.) is being recycled and disposed by auctioning through approved vendors only.	Implemented	E-waste and discarded equipment are collected in yellow bins and recycled through authorized vendors.
The excavated soil from various construction sites will either be reused at the same site or will be utilised for horticulture activities in different grounds.		
Waste audits will be performed on regular basis to understand the performance of existing practices and taking appropriate measures for further improvement.	In Progress	PHO digitalisation is in progress and regular discussions in the clean green happy campus committee are being held to improve the system

Event:

Seminar on Battery Reuse, Repurpose & Recycling

Body:GESH

Details:

The IIT Bombay Research Hub for Green Energy and Sustainability, in collaboration with the Department of Metallurgical Engineering and Materials Science, organized a one-day seminar on ‘Battery Reuse, Repurpose & Recycling’. Held in partnership with MRAI, IIT Madras School of Sustainability, and supported by bodies like the Battery Research Society of India and RMI, the seminar marked the beginning of a series on Circular Economy. It brought together stakeholders from academia, industry, and regulatory bodies to address challenges and opportunities in battery recycling, in line with India’s EV R&D roadmap. Expert talks covered policy, technology, and research directions, with speakers from MeitY, CPCB, PSA’s office, industry, and international academia

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Event:
Paper Collection Drive
Body:
NSS
Details:
The NSS paper collection drive at IITB was conducted from March 24 to March 28, 2025. This recurring initiative encouraged students, faculty, and staff to donate used papers, answer sheets, and newspapers by placing them in collection boxes located across hostels, academic areas, and department buildings. The drive promoted environmental responsibility and sustainability by ensuring collected paper was recycled, thereby reducing campus paper waste. It contributed to a cleaner, greener campus and raised awareness on sustainable practices, forming a key part of NSS’s broader programs in ecological conservation and community service at IITB

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11. Energy conservation

Goals	Progress Icon	Details
Increasing the contribution of solar photovoltaic power will be one of the main energy-saving measures.	In Progress	We have a large solar energy plant above VMCC. Also numerous solar plants implemented in various department buildings
Some examples for energy conservation may include replacement of CFL by LED, use of motion sensor activated lights and exhausts in toilet blocks and installation of BLDC fans. etc.	Completed	A significant proportion of CFL bulbs are replaced by LEDs, and motion sensor activated lights are implemented in multiple washrooms across hostels
Further, the new buildings will be constructed in such a way that the natural light is sufficient in day time.	Completed	The feature is efficiently implemented in H7, H8 and H21 (Project Evergreen)
Energy audits will be performed periodically for continual improvement in	In Progress	GESH and IEAC are in progress of drafting a proposal for the project

energy systems and their usage.		

Event:

Solar Lamp Making Workshop

Body:

Energy Club, Team Shunya

Details:

Team Shunya and the Energy Club, together with NGO GNAAN U, hosted a workshop where students and community members learned how to build solar lamps. Working in small groups and guided by mentors, participants assembled lamps using simple, easily available materials. The focus was not just on the technical steps but also on understanding why clean energy access matters, especially in places where electricity isn’t always reliable. The group took time to discuss real challenges around energy use and the environment, making the experience more meaningful. Through demonstrations and open conversation, everyone gained a clearer picture of how solar technology can improve lives and contribute to sustainability. The event gave people practical skills and sparked new interest in science and social impact,showing how renewable energy can be both accessible and empowering

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Event:

Workshop on Solar OASIS Project with SUNRISE

Body:

GESH

Details:

The Solar OASIS workshop, hosted by IIT Bombay in collaboration with SUNRISE, brought together researchers, community members, and experts from across India and the UK to discuss how solar-powered infrastructure can support rural development. Held in Khuded village, the workshop focused on the Solar OASIS building, a community space that runs entirely on solar energy and supports services like lighting, milling, cooling, and phone charging. Through open discussions and breakout sessions, the event explored what worked, what could be improved, and how similar models could be adapted for other rural areas. It was a grounded and collaborative effort to bridge sustainable technology with real community needs.

Event:

Green Hydrogen Reality Check 2025

Body:

GESH

Details:

The IIT Bombay Research Hub for Green Energy and Sustainability, supported by HSBC, is hosting Green Hydrogen Reality Check 2025 on 20–21 Feb 2025 at VMCC, IIT Bombay. The two-day conference will focus on green hydrogen technologies, policy, and finance, aiming to bridge the gap between vision and reality. With sessions on technology adoption, governance, and industry-academia collaboration, the event will serve as a key platform to drive India’s green hydrogen transition.

Event:

Renewathon: Real World Challenge in Renewable Energy

Body:

Energy Club

Details:

Organized by the Energy Club in collaboration with Fourth Partner Energy, it was a hackathon focused on solving real-world renewable energy challenges. Open to IIT students, it promoted innovation and industry engagement in clean energy. The competition encouraged teams to address practical issues provided by industry experts, fostering solutions with potential real-world application. Top 2 ranked teams got the opportunity to intern at FPE. The club’s 2025–26 Power Squad responsible for organizing major sustainability initiatives, including Renewathon was also introduced through this. The event underscored the importance of hands-on learning, industry collaboration, and proactive student participation in India’s energy transition efforts.

Event:

En-ROADS Climate Action Simulation Workshop

Body:

Energy Club and GESH

Details:

The Energy Club in collaboration with ClimatexCapital, organized a series of En-ROADS simulation challenges, engaging students in dynamic, interactive workshops on climate policy and energy transition. Using the En-ROADS climate solutions simulator, participants explored various scenarios for reducing global greenhouse gas emissions and limiting global warming. The sessions were led by expert facilitators and included group discussions, policy debates, and hands-on experimentation with the simulator. The challenges not only deepened participants’ understanding of climate science and energy systems but also empowered them to think critically about real-world solutions. The Energy Club’s initiative fostered a sense of urgency and agency among students, inspiring them to take an active role in addressing the climate crisis.

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12. Sustainable infrastructure development

Goals	Progress Icon	Details
Its features include energy and water saving fixtures, proper ventilation (i.e., air exchange with the outside environment), and natural lighting.	In Progress	The features are efficiently implemented in H7, H8 and H21 (Project Evergreen)
The new buildings will be built to target	In	A 3 Star (mandatory) rating is required.

GRIHA four-star rating criteria.	Progress	H7, H8 and H21 are 4 star GRIHA Certified.
Renewable energy through wind and solar PV may be installed and green power will be wheeled for campus use.		
The new construction will be planned in a manner to reduce land footprint and conserve greenery.	In Progress	The objectives are being followed in context of GRIHA Certification

Event:

Sustain Talk: Demystifying Green Building Rating Systems

Body:

Team Shunya

Details:

"SUSTAIN Talks" is a public lecture series included in Sustain 4.0 hosted by Team SHUNYA, the IIT Bombay student group focused on sustainable housing, net-zero solar-powered homes, and broader sustainability and energy topics

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Event:

Sustain Talk: Demystifying Green Building Rating Systems

Body:

Team Shunya

Details:

Project Vivaan is a net positive energy, net zero water, and net zero carbon sustainable housing prototype designed for warm, humid climates. It features advanced construction, renewable energy integration, water-saving fixtures, and automation. Recognized at the Solar Decathlon Build Challenge, it involved major industry collaborations with Autodesk and others, incorporating Digital Twin modelling. These partnerships enabled real-world testing, funding, and scalability for sustainable, affordable urban housing in India.

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13. Road cleaning & green cover and landscaping

Goals	Progress Icon	Details
Suitable measures will be taken to prevent dust suspension in air during road cleaning in the campus. Hence, vehicle mounted road sweeper having a provision for water spray will be used for cleaning of the roads.	In Progress	Dust Guns are currently used to prevent dust suspension and cleaning is done everyday.

The Institute will make all efforts to conserve greenery in the campus by compensatory tree plantation, growing grass and native shrubs.	In Progress	Institute Bodies like primarily Horticulture (through DEAN IPS), NSS, PRO are doing this.
In addition, more playfields will be developed for students, staff and family members. Kid's parks will also be developed at various locations.	Completed	There are 2 parks in the institute - near Hostels 15 and 17.

Event:

Versova Beach Cleanup Campaign (Abhyuday)

Body:

Abhyuday

Details:

Abhyuday, the social body of IIT Bombay, actively participates in the Versova Beach Cleanup alongside Afroz Shah's team. Held monthly, this initiative fosters environmental responsibility among students while contributing to the restoration of Mumbai's coastline. It highlights the power of youth driven volunteerism in combating urban pollution and promoting sustainability.

Event:

Flare - River Cleanup Drive @ Mithi River (NSS)

Body:

NSS

Details:

During the River Cleanup Drive at Mithi River, NSS IIT Bombay extended its impact beyond environmental conservation by organizing a food drive for local communities. Volunteers distributed nutritious meals to underprivileged individuals residing near the riverbank, many of whom face food insecurity on a daily basis. The initiative was not only about providing immediate relief but also about raising awareness of the link between environmental health and social well-being. By engaging with local residents, NSS volunteers gathered insights into the challenges faced by marginalized communities and explored ways to address both ecological and humanitarian needs. The food drive became a symbol of solidarity and compassion, demonstrating how sustainability efforts can be intertwined with social responsibility and community service.

Event:

Advitiya - Powai Lake Cleanup Drive (Abhyuday)

Body:

Abhyuday

Details:

The drive took place on January 11, 2025, starting at 7 AM, with volunteers gathering to restore and clean a significant portion of Powai Lake within the IIT Bombay campus. The campaign successfully mobilized the community to clean a substantial 2km stretch of the lake's shoreline, focusing on removing waste and promoting environmental responsibility. The event highlighted collective action for ecological conservation and aimed to inspire ongoing efforts for a cleaner, greener campus and local environment.

Event:

Mangrove Cleanup Drive at Coastal and Marine (NSS)

Body:

NSS

Details:

The Mangrove Cleanup Drive at the Coastal & Marine Biodiversity Centre in Airoli was organized by NSS IIT Bombay in partnership with Beach Please. Volunteers collected and segregated waste from the mangrove area, supporting conservation and raising awareness about the importance of healthy coastal ecosystems. Certificates and refreshments were provided, making the drive both meaningful and engaging for all participants.

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14. Biodiversity management

Goals	Progress Icon	Details
Biodiversity audits will be conducted periodically.		
IIT Bombay has created a protected no-construction zone next to the Powai lake, protecting the natural flora and fauna both on land, and in the wetlands leading to the lake.	Completed	The plan is already implemented and being monitored.

Event:

Blooming Hands – Plant Care Initiative (NSS)

Body:
NSS
Details:
The Blooming Hands – Plant Care Initiative by NSS IIT Bombay promotes environmental consciousness through engagement with nature. Volunteers regularly care for plants across campus, ensuring their growth and maintenance while beautifying shared spaces. The initiative instills a sense of responsibility toward green living and sustainability among students. By nurturing flora within the institute, Blooming Hands serves as a quiet yet impactful reminder of our duty to preserve the environment, one plant at a time.

Event:
Tulsi Mega Plantation Drive (Team Shunya & NSS)
Body:
Team Shunya
Details:
The Tulsi Mega-Plantation Drive, held on 26th January by NSS IIT Bombay and Team SHUNYA, brought together the campus community to plant tulsi saplings, enhancing green cover and fostering awareness about sustainable living. This initiative promoted environmental stewardship and encouraged participants to take an active role in nurturing a healthier ecosystem through organic plantation efforts. The drive not only beautified the campus but also reinforced a collective commitment to sustainability and responsible interaction with nature.

Event:
Tree Plantation in Khopoli (NSS)
Body:
NSS
Details:
NSS IIT Bombay, with The Recycling Company, organized a tree plantation drive in Khopoli. Volunteers planted native species in a designated area, contributing to reforestation and habitat restoration. The event included educational sessions on the importance of trees for climate resilience and biodiversity. By working with local partners, NSS extended its impact beyond the campus and fostered a sense of shared responsibility for the environment.

Event:
Vrikshabandhan – Tying Rakhis to Trees (NSS)
Body:
NSS
Details:
NSS IIT Bombay organized Vrikshabandhan events where volunteers tied rakhis to trees on campus, pledging to protect and nurture them. The activities aimed to promote environmental responsibility, raise awareness about tree conservation, and strengthen campus sustainability through group gatherings and symbolic actions.

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15. Handling, storage and disposal of hazardous materials and wastes

Goals	Progress Icon	Details
The hazardous chemicals or materials used in research activities should be stored in the appropriate material storage cabinets. Proper ventilation should be provided wherever necessary.	Completed	It's already being done by the Public Health Office (PHO) and the system is monitored regularly.
The proper disposal of hazardous liquid and solid waste will be ensured. Proper storage of waste should be the responsibility of waste generators.	Completed	It's already being done by the Public Health Office (PHO) and the system is monitored regularly.
These types of wastes have to be stored in leak proof containers with all precautions before handing over to the vendor. Suitable labels should be put on the containers to prevent any mishandling.	Completed	It's already being done by the Public Health Office (PHO) and the system is monitored regularly.

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16. Improved Housekeeping Activities

Goals	Progress Icon	Details
Use of mechanized equipment and ensuring avoidance of direct human contact with solid and liquid waste will not only improve the cleanliness but also protect the health of housekeeping workers.	In Progress	This is being done at certain locations, others will be implemented soon.
For their safety and health, the workers will be provided the personnel protective equipment (PPE).	Completed	This was efficiently being done during the times of high risk (e.g. Covid Pandemic).

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17. Safety Aspects

Goals	Progress Icon	Details
The Fire & Safety Section (FSS) of IIT Bombay will prepare a comprehensive document containing Standard Operating Procedures (SOPs) for various types of safety such as lab safety, hospital safety, hostels & residences safety, road safety and construction safety.		
The students will be educated through sessions to be organized by Safety Officer, FSS.	Completed	This is being done in Hostels. Fire Safety awareness quiz was conducted on ASC. A fire safety session is conducted before labs.
Mock drills will be performed at least once in a year to train volunteer staff and students.	Completed	Being done in Hostels. Fire education drills are demonstrated in all hostels.
Internal safety audits will be conducted twice a year whereas external audits will also be conducted once in two years.		

Event:
Health Camp
Body:
NSS
Details:
NSS IIT Bombay, in collaboration with Aamra Care, organized a comprehensive health camp focused on the well-being of construction and PHO (Physical Health Office) workers on campus. The camp was designed to provide accessible healthcare services to a group that is often overlooked in institutional wellness programs. Volunteers facilitated the smooth operation of the camp, offering medical check-ups, blood pressure and glucose monitoring, and consultations with healthcare professionals. Workers were also educated on occupational health and safety, with sessions on proper ergonomics, injury prevention, and mental well-being. The initiative fostered a sense of inclusion and care, highlighting the importance of holistic health for all members of the campus community. By addressing both physical and mental health needs, NSS reinforced its commitment to building a healthier, more supportive environment for everyone.

Event:
FLARE 2024 (Blood Donation, Food Drive) (NSS)

Body:

NSS

Details:

The Blood Donation Drive organized by NSS IIT Bombay on September 28, 2024, at the Convocation Foyer encouraged students and staff to donate blood in partnership with RK Copper Blood Center. The drive highlighted the importance of blood donation in saving lives and fostered a sense of community responsibility among participants.

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18. Governance

Green Cup

What is it?

The Green Cup is the flagship initiative of IIT Bombay's Sustainability Cell, a yearly inter-hostel competition that promotes sustainability through friendly rivalry. It encourages hostels to adopt greener practices in areas like electricity use, water conservation, waste management, and awareness events.It concerns the hostel residents as it mainly helps in waste reduction.

Each hostel is ranked using the Green Cup Index, which measures sustainability across waste, electricity, and water. This year, the index has been updated to include new criteria like installation of meters and systems for regular data collection helping hostels build long-term eco-friendly infrastructure.

Primary Stakeholders involved:

1.Hostel Council (HC): Implements sustainability initiatives,coordinates hostel-level activities,ensures compliance with Green Cup criteria.

2.Sustainability Secretaries (Under HC):Leads hostel-specific sustainability projects, raises awareness among students, facilitates Green Cup participation.

3.Mess Council & Caterers: Ensures food waste management,implements segregation policies, tracksmess-related sustainability metrics.

4.Public Health Office (PHO):Oversees waste management,sanitation, and hostel cleanliness,provides data for Green Cup assessment.

How did it come into being?

The Green Cup started as an effort to increase awareness and action around sustainability in the hostel ecosystem. The idea was to make sustainability fun, competitive, and something every student could be part of.

Over the years, certain gaps were noticed; unclear roles, lack of coordination, and missing data. So, from this year:

- Working on creating on ground data collection pipelines digitally for waste with attached proofs for credibility
- A new hostel-level position, Sustainability Secretary, has been introduced to lead Green Cup activities and coordinate with the Sustainability Cell.
- A detailed SOP Booklet has been shared to guide Sustainability Secretaries and hostel councils on what actions to take and how to implement them.

These changes aim to make participation easier, smoother, and more impactful.

<u>What will be evaluated?</u>

Hostels will be scored on:

- Mess food wastage (20%)
- Mess food segregation (15%)
- Hostel waste management (10%)
- Electricity usage (30%)
- Zero wastewater discharge (10%)
- Attendance at the Green Cup launch (5%)

Most of the data will be sourced from the Public Health Office and Electrical & Maintenance Division, and backed by digital records.

<u>What's new this year:</u>

- There will also be an impact report and proof in a fixed format to be filled by hostel councils for every initiative they take up a revised scoring system with multiple levels that rewards hostels for consistent and proactive efforts.
- New points for setting up meters and systems to track water and electricity data.
- Regular digital data collection to ensure records are up-to-date and verifiable.

These changes help promote accountability, data-driven decisions, and future-readiness.

<u>How it's going:</u>

Hostels are now better equipped to take on the Green Cup, with a clearer structure, stronger leadership, and tools that reward long-term sustainability.

With Sustainability Secretaries in charge, detailed SOPs in place, and a new focus on real initiatives and data tracking, the Green Cup is becoming more effective each year.

The aim isn't just to win,it's to build a campus-wide culture of responsibility, climate action, and community driven change.

Previous Year's Leaderboard (as of March 23, 2025)

Event:

Carbon Footprint Calculator

Body:

Team Shunya

Team:

The Carbon Footprint Calculator developed by Team SHUNYA, IIT Bombay, empowers individuals and campus stakeholders to understand and reduce their environmental impact. By collaborating with faculty experts, including Professors Vikram Vishal and Arnab Dutta, the team conducted a detailed campus-wide carbon audit, identifying that around 95% of emissions stem from electricity use (Scope 2), with minimal shares from Scope 1 and Scope 3. This tool provides a practical benchmark for sustainable planning and actionable insights to minimize carbon footprints through informed policy and behavioural change.

Event:

Guest Lecture on enabling sustainability business by science

Body:

GESH

Team:

The Guest Lecture on “Enabling Sustainable Business by Science” hosted at IIT Bombay’s Department of Chemical Engineering in collaboration with GESH IITB featured Dr. Carla Seidel, Senior VP of Analytical & Material Science at BASF, Germany, who shared real-world insights on how scientific innovation drives sustainable business practices in the global chemical industry. Held on 13 December 2024, the session explored circular economy models, material science breakthroughs, and evidence based strategies to accelerate green growth across sectors.

Event:

Winter Projects: Corporate Sustainability & Sustainable Investing

Body:

Sustainability Cell

Details:

The Winter Projects 2024 by the Sustainability Cell, IIT Bombay, offered students hands-on experience in two key areas: Corporate Sustainability and Sustainable Investing.

In the Corporate Sustainability track, students explored major ESG reporting frameworks like MSCI, Sustainalytics, CDP, and BRSR, while learning about double materiality, stakeholder mapping, and KPI design. The project culminated in drafting an ESG policy for a chips manufacturing firm, covering supply chain emissions and sustainability strategy.

In the Sustainable Investing track, students examined how Venture Capital, Private Equity, and Green Bonds drive ESG goals. Tools like the Morningstar Sustainability Rating and case studies (e.g., Proterra, Beyond Meat) guided them in building a sustainable $3M investment portfolio with a cost-benefit analysis.

Together, the projects bridged ESG theory with real-world application in business and finance.

Event:

Impact Innovators Summit: CSR & ESG 2024

Body:

Abhyuday

Details:

The event brought together CSR and ESG experts, NGO leaders, and industry stakeholders for panel discussions, networking, and deliberations on integrating CSR and ESG into sustainable strategies. The summit promoted collaboration, innovation, and scalable impact through knowledge-sharing and strengthened IIT Bombay's leadership in sustainability and social responsibility across sectors. Participants explored how innovation and governance can drive lasting change, with actionable insights shared for enhancing corporate sustainability frameworks. The event also reinforced the importance of cross-sector partnerships in addressing complex environmental and societal challenges.

Event:

Annual CSR Conclave

Body:

IITB

Details:

The IIT Bombay Annual CSR Conclave 2025, themed "Fuelling Technology for Nation Building," brought together over 500 industry leaders, academics, and development professionals to advance societal impact through technology-driven partnerships. As covered by the Times of India, the event featured the launch of the Annual CSR Catalogue 2025 with 276 actionable projects, showcased major collaborations with over 140 companies, and included focused panel discussions, lab tours, and a felicitation ceremony celebrating top CSR contributors, reinforcing IIT Bombay's role in driving national development and innovation.

Event:

Workshop on Incorporating Sustainability in UG ChemEng Curriculum

Body:

GESH

Details:

In the one-day workshop, more than 30 academic and industry participants engaged in a panel discussion, two break-out sessions on course revision and case studies, and collaborative brainstorming. Led by Prof. Yogendra Shastri, the workshop generated ideas to develop new teaching materials and strengthen industry-academia cooperation for sustainability education. Additional contents covered included discussions on the industry's expectations from chemical engineering graduates regarding sustainability competencies, and reviews of existing course material to identify opportunities for integrating real-world sustainability case studies into the curriculum

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Event:

QS Rankings

Body:

GESH

Details:

GESH IITB took the responsibility of completing the sustainability baseline data collection and analysis of IITB, this was submitted for QS World Rankings. IIT Bombay has made a remarkable leap in the ranking for Sustainability, improving its score from 52.5 in 2025 to 75.2 in 2026. This achievement reflects the collective commitment of faculty, staff, students, and campus residents to embed sustainability across the institute. GESH, in collaboration with the Dean IPS Office, supported the Dean Strategy Office in data collection and reporting for the ranking. The improvement highlights the campus-wide adoption of sustainable practices—from energy and waste management to curriculum and community engagement

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Event:

STARS rankings

Body:

Sustainability Cell & GESH

Details:

The Sustainability Cell along with GESH conducted IIT Bombay's first comprehensive assessment under the STARS 3.0 framework, analyzing over 500 parameters across academics, operations, and community engagement. Currently the team is performing preliminary analysis while identifying areas for improvement.

Event:

Conference of Youth India 2024 (ZeroWaste)

Body:

Zero Waste

Details:

Team Zero Waste, IIT Bombay partnered as a knowledge collaborator for LCOY India 2024, actively promoting youth-led sustainability and climate action. They invited Indian youth to contribute policy recommendations on critical themes such as climate adaptation, finance, human rights, health, and innovation, directly influencing the National Youth Statement and international climate negotiations. Through workshops and consultations, Team Zero Waste empowered youth participation in decision-making while advancing practical solutions in waste management and sustainability for broader impact at both national and global levels.